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5. Drinking water contains a number of ions which are important in the human body. Some of these ions cause hardness in water.

The table shows the concentration of ions in drinking water from four different locations.

Location	Concentration of ions (mg/dm^3 of water)					
	potassium K^+	ammonium NH_4^+	calcium Ca^{2+}	fluoride F^-	sulfate SO_4^{2-}	nitrate NO_3^-
A	0.1	0.4	0.0	0.0	0.4	0.2
B	0.0	0.3	0.4	4.4	0.2	0.0
C	0.2	0.6	2.7	0.4	0.0	0.1
D	3.4	2.1	1.0	2.1	2.5	2.3

(a) Which location is likely to have the hardest water?

Put a tick (✓) in the box next to the correct answer. [1]

A ☐

B ☐

C ☐

D ☐

(b) In which location do people have the least protection against tooth decay from their drinking water?

Put a tick (✓) in the box next to the correct answer. [1]

A ☐

B ☐

C ☐

D ☐

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(c) Ionic compounds dissolve in water to form positive and negative ions.

Which **two** compounds may have dissolved in the drinking water at location **B**?

Put a tick (✓) in the box next to the correct answer. [1]

potassium fluoride and calcium sulfate ☐

ammonium sulfate and potassium nitrate ☐

calcium fluoride and ammonium nitrate ☐

ammonium sulfate and calcium fluoride ☐

(d) (i) Calculate the relative formula mass (M_r) of calcium sulfate, $CaSO_4$. [1]

$A_r(Ca) = 40$ $A_r(S) = 32$ $A_r(O) = 16$

M_r = ...

(ii) Calculate the percentage by mass of fluorine in calcium fluoride, CaF_2. [2]

$M_r(CaF_2) = 78$ $A_r(F) = 19$

Percentage = ... %

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